

CBSE Class 10 Mathematics — Circles

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains *questions. All questions are compulsory.*
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If the angle between two tangents drawn from a point P to a circle of radius r and center O is 60° , then the length of OP is:
(A) (A) r
(B) (B) $2r$
(C) (C) $r\sqrt{3}$
(D) (D) $r/2$
2. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. The length of PQ is:
(A) (A) 10 cm
(B) (B) 8 cm
(C) (C) 12 cm
(D) (D) 7 cm
3. The number of tangents that can be drawn from a point lying inside a circle is:
(A) (A) 0
(B) (B) 1
(C) (C) 2
(D) (D) Infinite
4. If two tangents inclined at an angle of 80° are drawn to a circle of radius 3 cm, then the angle between the two radii at the center is:
(A) (A) 80°
(B) (B) 90°
(C) (C) 100°
(D) (D) 120°

5. The length of a tangent from a point A at a distance of 10 cm from the center of the circle is 8 cm. The radius of the circle is:
- (A) (A) 4 cm
 - (B) (B) 6 cm
 - (C) (C) 7 cm
 - (D) (D) 9 cm
6. Assertion (A): The length of a tangent drawn from an external point to a circle is equal to the length of another tangent drawn from the same point. Reason (R): The lengths of tangents drawn from an external point to a circle are equal.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): The tangent at any point of a circle is perpendicular to the radius through the point of contact. Reason (R): The perpendicular distance from the center of the circle to the tangent is equal to the radius.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A point inside a circle lies in the interior, and any line passing through it will be a secant, not a tangent.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): If the angle between two tangents drawn from a point P to a circle of radius r and center O is 60° , then $OP = 2r$. Reason (R): The line segment from the center to the external point bisects the angle between the tangents.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
10. Assertion (A): The length of the tangent from a point A at a distance of 5 cm from the center of the circle is 4 cm. The radius of the circle is 3 cm. Reason (R): In a right triangle formed by radius, tangent, and hypotenuse (distance to center), $r^2 + t^2 = d^2$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. Find the length of PQ .
2. If tangents PA and PB from a point P to a circle with center O are inclined to each other at an angle of 80° , then find $\angle POA$.
3. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
4. A quadrilateral $ABCD$ is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, find the length of AD .
5. If a point P is 26 cm away from the center O of a circle and the length of the tangent PT drawn from P to the circle is 10 cm, find the radius of the circle.

Section D: Long Answer (4-5 marks each)

1. A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.
2. From a point Q , the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. Find the radius of the circle.
3. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
4. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80° , then find $\angle POA$.
5. Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line segment joining the points of contact at the centre.